Report

Task 1

This task introduces us to the Keras API, and shows us how to inspect the details of the image thats been loaded. It also shows how to show the image.

```
<class 'PIL.JpegImagePlugin.JpegImageFile'>
JPEG
RGB
(1920, 2879)
```

Task 2

This task shows us how to convert the image to and from the numpy array, and then inspect the numpy arrays details.

```
<class 'PIL.JpegImagePlugin.JpegImageFile'>
float32
(2879, 1920, 3)
<class 'PIL.JpegImagePlugin.JpegImageFile'>
```

Task 3

This task shows us how to change the colour scale of an image and how to save it, using the os builtin library.

```
<class 'PIL.Image.Image'>
None
RGB
(1920, 2879)
```

Task 4

This task introduces us to the Keras test datasets and how to calculate their statistical summaries in regards to pixel values, and their dataset shapes.

```
Train (60000, 28, 28) (60000,)
Test ((10000, 28, 28), (10000,))
Train 0 255 33.318421449829934 78.56748998339798
Train 0 255 33.791224489795916 79.17246322228642
```

Task 5

This task basically makes the dataset be grayscale (reducing the color channels to 1) and then gets the scale of the images available in the dataset, generates a rescale amount, sets up the images to be iterated over to rescale them and confirms that worked.

```
Train min=0.000, max=255.000
Test min=0.000, max=255.000
Batches train=938, test=157
Batch shape=(64, 28, 28, 1), min=0.000, max=1.000
```

Task 6

This task basically does the same but it does it for the mean/average pixel value across the dataset.

```
Means train=33.318, test=33.791
Data Generator Mean: 33.318
(64, 28, 28, 1) 0.25400496
(60000, 28, 28, 1) 1.1267007e-06
```

Task 7

This one does the same, but it centres the pixel values instead as far as I can tell.

```
Statistics train=33.318 (78.567), test=33.791 (79.172)
Data Generator mean=33.318, std=78.567
(64, 28, 28, 1) -0.019072345 0.97481656
(60000, 28, 28, 1) 1.3452647e-08 1.0
```